

JKBOSE 2026

CLASS 11TH

Photosynthesis in Higher Plants

LECTURE-02

BOTANY

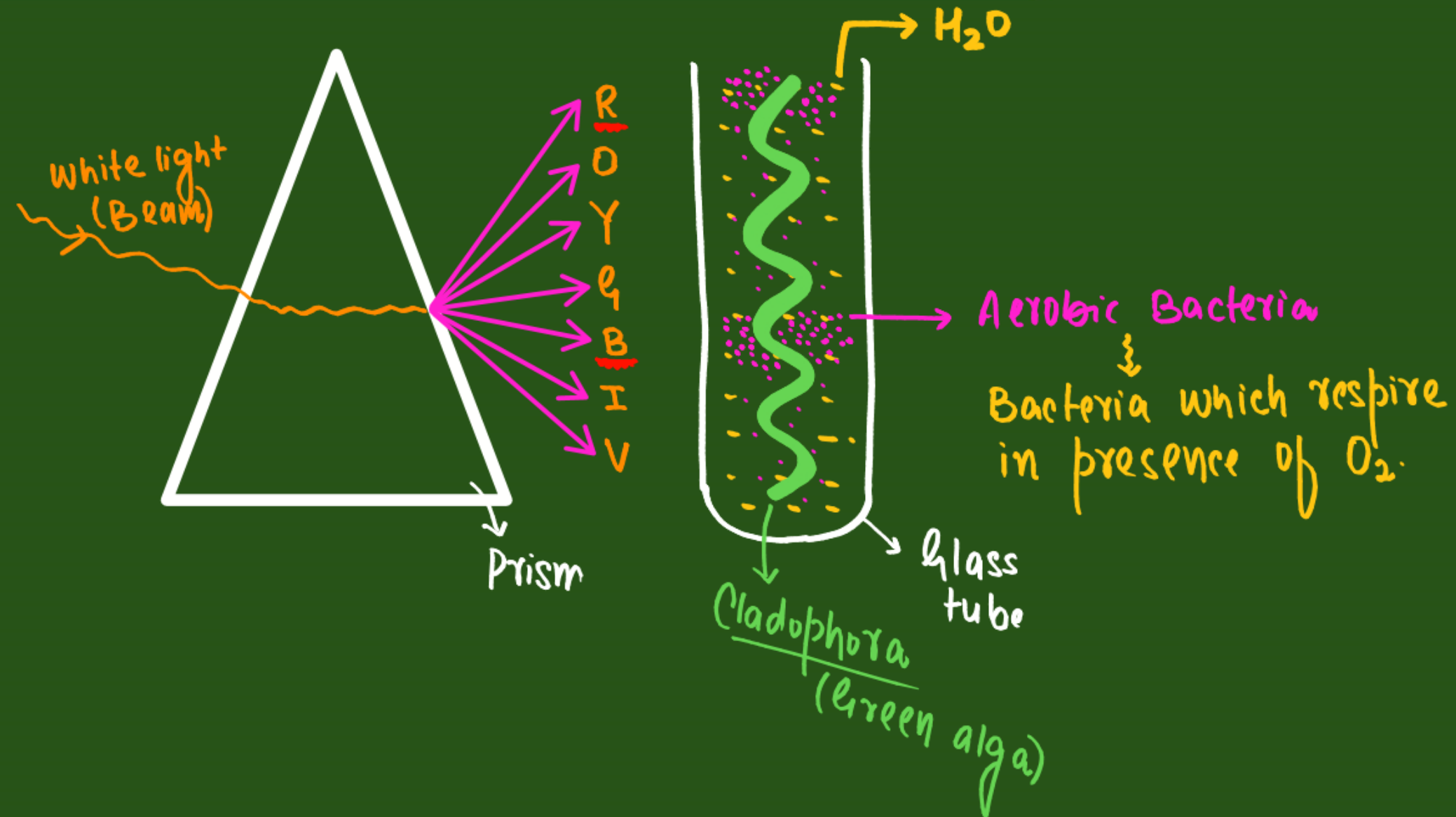
ARSHID BHAT SIR



Topics To Covered

- 1** Experiments Related To Photosynthesis (contd..)
- 2** ...

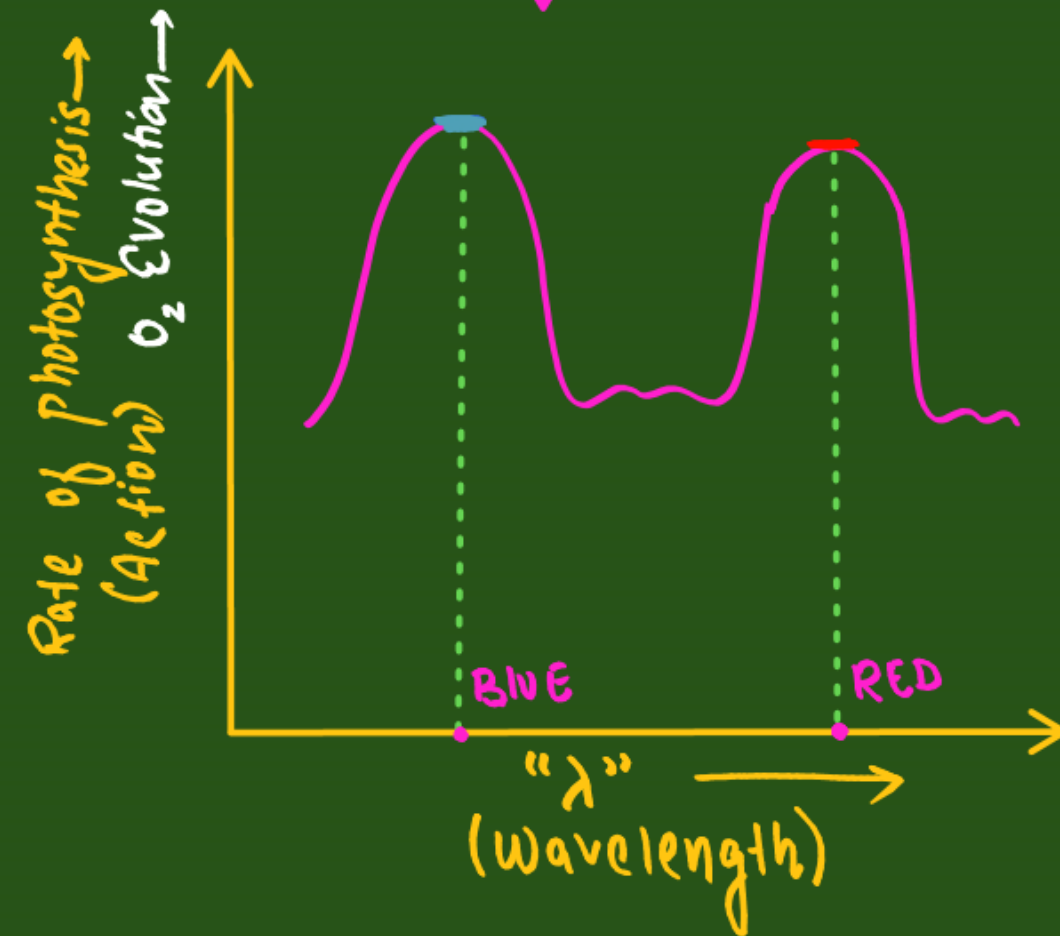
3. T.W Englemann (1843-1909)



- T.W Englemann - Split white light into its Spectral Component by using glass prism.
- He illuminated a green alga (Cladophora) into a suspension of aerobic bacteria.
- He observed accumulation of aerobic bacteria in the regions of Blue and Red.
- He used aerobic bacteria to detect sites of O_2 evolution.
- He observed effect of different wavelengths of light on rate of photosynthesis.

Blue and Red region - maximum photosynthesis.

≡ T.W Englemann first time draw **Action Spectrum** for photosynthesis.



4. Julius Von Sachs (1854)

- ≡ He provided evidence for production of glucose when plants grow.
- ≡ He also confirmed that this glucose gets stored as **Starch**.
- ≡ His later studies also confirmed that:
 - a) In plants, there is **green structure** (now called as chloroplast).
 - b) Inside this green structure is present a **green chemical** (now called as chlorophyll).

INFERENCE Production and storage of glucose in plants.

5. Cornelius Von Neil (1897-1985)

- ≡ He was a microbiologist.
- ≡ He provided a milestone contribution in understanding the phenomenon of photosynthesis.
- ≡ He demonstrated that, photosynthesis occurs in two steps:

a) Light Reaction:

- Here splitting of H_2O takes place.
- Main role of H_2O is H-donor.



b) Dark Reaction:

- Here CO_2 gets reduced to form $C_6H_{12}O_6$.



6. Ruben and Kamen

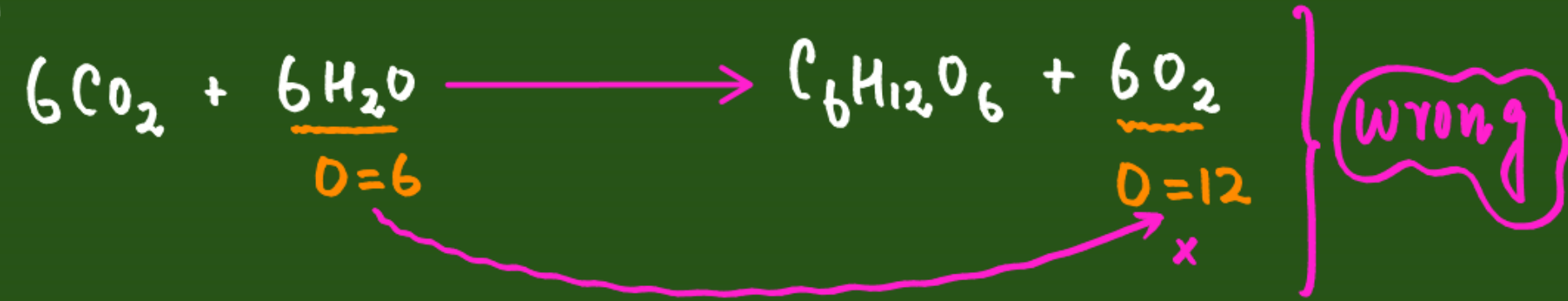
- ≡ They confirmed that, O_2 evolved during photosynthesis comes from H_2O and not from CO_2 .
- ≡ They used **radio-isotope labelling technique**.

• O_2 $\left\{ \begin{array}{l} \text{Normal isotope } (O^{16}) \\ \text{Radioactive isotope } (O^{18}) \end{array} \right.$

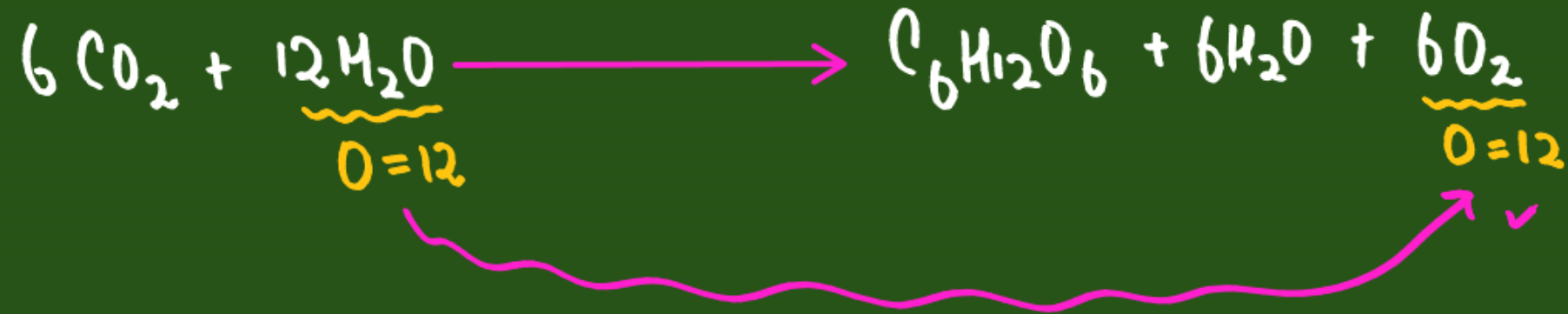


NOTE:

- By the Middle of 19th, the Equation of photosynthesis was understood as:



- The Correct Equation of overall process of photosynthesis is:



QUESTIONS

Who used prism, white, light green alga Cladophora, and aerobic bacteria and plotted the first action spectrum for photosynthesis?

- A** Sachs
- B** Arnon
- C** Arnold
- D** Engelmann

QUESTIONS

liberated Oxygen which is during photosynthesis comes from

- A** Carbon dioxide
- B** Water
- C** Chlorophyll
- D** Phosphoglyceric acid

QUESTIONS

The oxygen released during photosynthesis comes from water was proved with the help of ^{18}O experiment by

A Ruben and Kamen

B Hill

C Warburg

D Blackman

QUESTIONS

H/W



Name of the scientist who pointed out the importance of different wavelengths of light using a green algae and aerobic bacteria:

- A** Priestley
- B** Ingen-Housz
- C** KV Thimann
- D** Englemann

H/W

QUESTIONS

Select the mismatched pair w.r.t. scientists and their contributions in photosynthesis:

	Scientist	Contribution
A	Joseph Priestley	Discovered oxygen
B	Jan Ingenhousz	Showed that sunlight is essential for plant process
C	Julius von Sachs	Provided evidence for production of glucose when plants grow
D	Engelmann	Water is the source of O_2

HOMework

2 Questions





(5 Star For UR Hardwork)

Thank
You